

Madhan Raghuram

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SUMMARY

Software Developer with a strong interest in building **scalable** and **efficient systems**. Focused on understanding how applications work end to end and designing solutions that are simple and maintainable. Currently exploring **AI engineering** and working towards integrating intelligent, data-driven features into modern software systems.

EDUCATION

2023 – 2027 Chennai, India	Prince Dr K Vasudevan College of Engineering and Technology B.E Computer Science and Engineering (8.4 CGPA)
2022 – 2023 Chennai, India	Sacred Heart Matriculation Higher Secondary School Higher Secondary Education (83%)
2021 – 2022 Chennai, India	Sacred Heart Matriculation Higher Secondary School SSLC (Promoted under COVID assessment policy)

PROJECTS

Med-Bot (Hospital Query System) *🔗*, Python | LangChain | FastAPI | Streamlit | FAISS

- Developed a **RAG** application, implementing document ingestion, chunking, embedding generation and vector storage
- Engineered an LRU-based **caching system**, reducing repeated-query response latency from **~600ms to 15ms (97.5% improvement)** while lowering API costs.
- Architected an **admin panel** for efficient content updates, document uploads, and chat-log monitoring.

Schemify (Schema-Aware Test Data Generator) *🔗*, Python | FastAPI | Faker | Anthropic API

- Developed an AI-powered pipeline that parses SQL **DDL schemas** and generates realistic synthetic test data using intelligent LLM-assisted Faker mappings.
- Reduced LLM token usage by 90%+** by implementing schema **caching** and incremental column mapping, eliminating redundant API calls for **unchanged schemas** and improving cost efficiency.

BulkMailer (Automatic Bulk Emailer) *🔗*, Implemented for Galada Finance Limited

- Developed a **Python-based CLI tool** to automate **bulk email delivery** using Gmail SMTP and CSV integration, enhancing secure and personalized outreach.

PROFESSIONAL EXPERIENCE

Iproat Solutions, Chennai *🔗*, SDE - Intern | Jun 2025 – Jul 2025

- Assisted in developing **ChatBot** using **Laravel** and **OpenAI** to enable conversational access to employee information such as attendance, shift, and permissions.
- Optimized chatbot performance by routing common queries through **fuzzy string matching** and reserving OpenAI for complex interactions, reducing unnecessary **LLM API calls**.

SKILLS

Languages & Frameworks

Python, Java, C, SQL, React, Spring Boot, FastAPI.

Core Competencies

DSA, AI-Assisted Development (Claude Code, Github Copilot), Agile Development, API Development.

Databases & Cloud

AWS, Relational Databases (SQLite, MySQL), Non-Relational Databases (MongoDB), Vector Databases (FAISS, ChromaDB).

Tools & Platforms

Git, GitHub, Postman, Ollama.

CERTIFICATIONS

AWS Certified Cloud Practitioner *🔗*

- Validates a foundational understanding of core AWS services, security principles and operational best practices.

PERSONAL PURSUITS

E-Cell IIT Bombay, Campus Ambassador

- Promoted entrepreneurship** on campus through events, outreach, and **NEC** initiatives by E-Cell IIT Bombay.

LeetCode, Problem Solver

- Solved **170+** coding problems on **LeetCode** with a focus on Data Structures and Algorithms.

LANGUAGES

English | German – A1 | Tamil | Hindi – Level 4